

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 1: APPLICATIONS OF MATHEMATICAL KNOWLEDGE

Practice paper · Set 13

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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1 The roots of $2x^2 + bx + 18 = 0$ are equal. Find the sum of the possible values of b .

A 36

B 6

C 0

D 12

E 24

F -12

2 An arithmetic series has first term 5 and common difference 3. Find the sum of the first 40 terms.

A 2440

B 2540

C 1270

D 2600

E 2500

F 2580

- 3** A histogram has a bar for the class $10 \leq x < 15$ of height (frequency density) 3.2 and a bar for $15 \leq x < 25$ of height 1.5. What is the total frequency of these two classes?
- A** 15
- B** 23.5
- C** 31
- D** 4.7
- E** 16
- F** 47

4 Solve $\log_2 (x^2 - 4) - \log_2 (x - 2) = 3$.

A $x = 2$

B $x = 6$

C $x = -10$ or $x = 6$

D $x = 8$

E $x = 10$

F $x = 4$

- 5 For what range of values of x is $f(x) = 2x^3 - 15x^2 + 24x + 1$ decreasing?
- A** $-4 < x < -1$
- B** $2 < x < 3$
- C** $x > 1$
- D** $x < 1$ or $x > 4$
- E** $x < 4$
- F** $1 < x < 4$

6 £2000 is invested at 5% compound interest per year. Which expression gives the amount after n years?

A 2000×5^n

B $2000 + 100n$

C 2000×1.05^n

D 2000×1.5^n

E $2000(1 + 0.05n)$

F 2000×0.05^n

7 Find the minimum value of $x + \frac{4}{x}$ for $x > 0$.

A 5

B $2\sqrt{2}$

C 4

D 0

E 2

F 8

8 Class A has 20 students with mean mark 60; class B has 30 students with mean mark 70. What is the mean mark of all 50 students?

A 68

B 65

C 62

D 67

E 64

F 66

- 9** A recipe for 12 biscuits uses 150 g of flour and 100 g of butter. Ann has 1 kg of flour and 500 g of butter. What is the greatest number of biscuits she can make?
- A** 80
 - B** 60
 - C** 50
 - D** 48
 - E** 66
 - F** 72

10 For which values of x is $\sqrt{x^2} = -x$?

A $x = 0$ only

B $x \geq 0$

C $x \leq 0$

D $x < 0$ only

E all real x

F no real x

11 Find the value of $\frac{\log_5 27}{\log_5 3}$.

A 9

B $\frac{27}{3}$

C $\log_5 9$

D 3

E $\log_5 24$

F 24

12 Given that $\int_0^a (2x + 1) dx = 12$ where $a > 0$, find a .

A 4

B 12

C 2

D 3

E $\sqrt{12}$

F 6

13 A circle passes through $(0, 0)$, $(6, 0)$ and $(0, 8)$. Find the coordinates of its centre.

A $(5, 5)$

B $(2, 3)$

C $(4, 3)$

D $(6, 8)$

E $(3, 4)$

F $(3, 3)$

14 The line $y = x + k$ meets the circle $x^2 + y^2 = 8$ at exactly one point. Find the possible values of k .

A $k = \pm 2\sqrt{2}$

B $k = \pm 8$

C $k = \pm 4$

D $k = \pm \sqrt{8}$

E $k = 4$ only

F $k = \pm 2$

- 15** A line passes through $(1, 2)$ with gradient m . It crosses the x -axis at A and the y -axis at B , and the midpoint of AB is $(1, 2)$. Find m .

A -2

B $\frac{1}{2}$

C 2

D -4

E $-\frac{1}{2}$

F -1

- 16** A circle has radius r . A sector of angle θ radians has the same area as the segment cut off by a chord that subtends an angle of $\frac{\pi}{2}$ at the centre. Find θ .
- A** $\frac{\pi}{2} - 1$
- B** $\frac{\pi}{4}$
- C** $\frac{\pi}{4} - \frac{1}{2}$
- D** 1
- E** $\pi - 2$
- F** $\frac{\pi}{2} - \frac{1}{2}$

17 Find the complete set of values of k for which $x^2 + y^2 + 2kx - 4y + 3k + 4 = 0$ represents a circle.

- A** $k > 3$
- B** $0 < k < 3$
- C** $k < -3$ or $k > 0$
- D** $k < 0$ or $k > 3$
- E** $-3 < k < 0$
- F** all real k

- 18** The sum of the first n natural numbers is S . The sum of the first $2n$ natural numbers is kS . Which of the following is true?

A $k = \frac{4n+2}{n}$

B $k = 4$ for all n

C $k = \frac{2(2n+1)}{n+1}$

D $k = 2$ for all n

E $k = 4 - \frac{2}{n}$

- 19** Triangle PQR has area $10\sqrt{3}$, $PQ = 8$ and $PR = 5$. Given that angle QPR is obtuse, find QR .
- A** 13
- B** $\sqrt{89}$
- C** $\sqrt{89 + 40\sqrt{3}}$
- D** $\sqrt{129}$
- E** 7
- F** $\sqrt{109}$

- 20** The region bounded by $y = 4 - x^2$, the x -axis and the lines $x = 0$ and $x = 3$ is shaded. Find the total shaded area.

- A** 3
- B** $\frac{7}{3}$
- C** $\frac{16}{3}$
- D** $\frac{25}{3}$
- E** $\frac{23}{3}$
- F** 9

END OF TEST